

Development and Validation of a Novel Self-Report Measure of Monotropism in Autistic and Non-Autistic People: The Monotropism Questionnaire

Garau, V., Murray, A., Woods, R., Chown, N., Hallett, S., Murray, F., Wood, R. & Fletcher-Watson, S.

Valeria Garau, Department of Psychology, University of Edinburgh, ORCID: 0000-0003-4750-1315

Dr Aja Murray, Department of Psychology, University of Edinburgh, ORCID: 0000-0002-9068-3188

Richard Woods, School of Law and Social Sciences, London South Bank University, ORCID: 0000-0002-8292-632X

Dr Nick Chown, School of Law and Social Sciences, London South Bank University, ORCID: 0000-0001-6368-3249

Sonny Hallett, Autistic Mutual Aid Society Edinburgh, ORCID: 0000-0002-2667-6581

Fergus Murray, Autistic Mutual Aid Society Edinburgh, ORCID: 0000-0002-7770-9187

Dr Rebecca Wood, School of Education, University of Glasgow, ORCID: 0000-0001-8160-6212

Professor Sue Fletcher-Watson, Salvesen Mindroom Research Centre, University of Edinburgh. ORCID: 0000-0003-2688-1734

Abstract

Monotropism seeks to explain autism in terms of attention distribution and interests. Despite having strong subjective validity to autistic people, and potential to explain the overlap between autism and Attention Deficit Hyperactivity Disorder (ADHD), it has been little investigated formally. This is in large part due to lack of reliable and valid measures to capture the construct. In this study, we aimed to develop and validate a novel self-report measure, the Monotropism Questionnaire (MQ), in autistic and non-autistic people. The MQ consists of 47 items, which were generated by a group of autistic adults based on their lived experience and academic expertise. A sample of 1,110 participants (756 autistic, 354 non-autistic) completed the MQ online, as well as two established, short measures of autism, the Ritvo Autism Asperger Diagnostic Scale-14 (RAADS-14) and the Autism Quotient-10 (AQ-

10). Exploratory factor analyses suggested an 8-dimensional structure was optimal and the subscale scores showed high internal consistency reliability. The measure had very good convergent validity with the RAADS-14 and AQ-10 (all $r_s > .61$). Attention-deficit/hyperactivity disorder (ADHD) and autism status were both significantly associated with higher mean monotropism scores. Overall, the MQ had good psychometric properties. Future research should be undertaken to test this finding, particularly across demographic and diagnostic groups, and apply the MQ to predictions of Monotropism theory.

Keywords: attention, autism spectrum disorder, neurodiversity, ADHD, participatory research

Autism is diagnosed in the presence of a characteristic pattern of externally-observed features in two domains: *social communication and interaction* and *restricted and repetitive behaviours and interests* (APA, 2013). Although all autistic people¹ are diagnosed using highly similar clinical criteria, there is much variability in their experience (Happé & Frith, 2020). One driver of this variability is the fact that autism can co-occur with various other diagnoses, one of the most common being Attention Deficit Hyperactivity Disorder (ADHD), with the rate of ADHD in autism recently estimated at 38.5% (Rong et al., 2021). The overlap has been linked to atypical attention profiles, which are a central feature of both conditions (Murray et al., 2010). However, to date, there is no major, empirically-tested theoretical account of autism that centres attentional differences.

A potential explanation for the role of attention in autism is offered by Monotropism (Murray Lesser & Lawson, 2005), a theory of autism based on the lived experience of autistic people. It conceptualises the mind as an *interest-system*, meaning that individuals' interests guide their allocation of attention and different interests can be salient at different times. Here, *interest* is understood as anything that commands an individual's attention. Meanwhile, *attention* is conceptualised as a limited resource for which different cognitive processes are competing. Dinah Murray and colleagues (2005) posit that a monotropic mind has fewer interests aroused at any given time, with each one being allocated attention at a high concentration – this attention allocation strategy tends to apply to autistic people. In contrast, non-autistic individuals are hypothesised to be polytropic: they allocate attention at varying intensity levels to cover a broader range of interests, which may be co-aroused at any given time.

Little research has directly addressed monotropism theory despite its explanatory potential (Murray, 2018) but some investigations have taken place that specifically invoke monotropism as a mechanism. Monotropism may play a role in shaping how autistic people learn (Lawson, 2010; Leatherland, 2018; Wood, 2021). Monotropism has also been referenced in investigations of autistic sociality, that are often based on shared interests (Bertilisdottir-Rosqvist, 2019), while monotropic 'flow states' may be a key part of autistic wellbeing (McDonnell & Milton, 2014). At the same time, monotropism has been invoked to explain demand avoidance (Woods, 2019) – the same author predicts from a monotropism

¹ The language used to speak and write about autism and other neurotypes, especially in academic research, is still debated. In order to avoid perpetuating ableist ideologies and causing offence to the autism community, this paper adopts identity-first language and avoids using patronising and deficit-based language (see Bottema-Beutel et al., 2021).

account that autistic people should accrue trauma features over time, something that has been borne out in investigations of autistic susceptibility to PTSD (Rumball et al, 2021).

Monotropism has also been deployed as part of a cognitive difference-based account of ADHD (Bertilsson Rosqvist et al, 2023). Murray (2020) makes a number of further recommendations of how monotropism theory can and should be applied to understanding autistic people.

Despite the relatively small amount of direct invoking of monotropism as an explanation, the theory makes predictions about prominent features of autism that align with general research findings and subjective accounts from autistic people. For example, 75% to 95% of autistic people report having special interests (Turner-Brown et al., 2011), which are also enshrined in the diagnostic criteria (APA, 2013). Monotropism predicts the phenomenon of hyperfocus which research has repeatedly confirmed is a central aspect of autistic cognition, and of ADHD (Ashinoff & Abu-Akel, 2021; Grotewiel et al., 2022). Monotropism can also account for the well-documented difficulties many autistic people face when reorienting their attention, switching tasks, and processing multiple channels simultaneously (Goldstein et al., 2001; Collignon et al., 2013). Monotropism also aligns with observations of cognitive inflexibility in autism and ADHD (Craig et al., 2016) and strong emotional reactions to interruptions (Ozsivadijan et al., 2021). As a monotropic tendency makes simultaneously processing multiple channels difficult, this could explain why some of the channels involved in social interaction can go unregistered by autistic people (Collignon et al., 2013), or why the complex nature of school curricula can be inaccessible to some autistic pupils (Wood, 2021).

Theories of autism are important: they have the potential to enhance our understanding of autism, including the self-understanding of autistic people, to drive research hypotheses, and to suggest ways to improve the lives of autistic people. Consequently, it is crucial to ensure that theories of autism have face validity for autistic people, and the potential to deliver tangible benefits to them and their families. So far, the most dominant autism theories have not fulfilled these criteria (Happé & Frith, 2020). Monotropism is unique among cognitive theories of autism in its deployment of autistic expertise, its non-pathologising perspective, and its alignment with the neurodiversity paradigm (Milton, 2014). As academic psychology shifts towards connecting community priorities with rigorous scientific research (Fletcher-Watson et al., 2019), a psychometrically sound measure of monotropism would facilitate further empirical investigation of this theory.

The present research aims to develop and validate a novel self-report measure of monotropism in autistic and non-autistic people, using participatory research practices (Fletcher-Watson et al., 2019). We predicted significantly higher monotropism scores for autistic compared with non-autistic participants, and that the measure would show good convergent validity with established measures of autism. Since there is no background research to guide further hypotheses, the remaining analyses were exploratory. First, we explored whether ADHD status influenced the mean monotropism scores of autistic and non-autistic participants. Second, we investigated the existence of statistically and conceptually coherent sub-scales within the measure.

Methods

Ethics

Ethical approval was obtained from the ethics committee of the School of Philosophy, Psychology and Language Science at the University of Edinburgh (Ref. 233-2122/5). All participants provided informed consent.

Participants

Our recruitment strategy was informed by consultation with an autistic-led community group, and involved recruitment from the general population through social media, newsletters, and word-of-mouth. After excluding ineligible participants and those with fewer than 40/47 complete MQ items, our sample consisted of $n=1,110$ participants (756 classified as autistic, 354 classified as non-autistic) aged between 16 and 78 years ($M = 39$, $SD = 12$). The autistic participant group had a mean RAADS-14 score of 30.7 ($SD = 8.22$) and a mean AQ-10 score of 7.51 ($SD = 1.97$) and the non-autistic participant group had a mean RAADS-14 score of 10.3 ($SD = 9.02$) and a mean AQ-10 score of 3.31 ($SD = 2.31$). Further demographic characteristics are provided in Table S1 of Supplementary Materials.

Measures

Monotropism Questionnaire

A novel 47-item monotropism questionnaire was developed (see <https://osf.io/wpx5g/>) with item contents generated and iterated in collaboration with an autistic-led community group, including both academics and lay members with a shared interest in Monotropism Theory. This was based on their professional and personal experience of monotropism and

iterated across multiple rounds. Members first proposed items and commented on other members' item suggestions. Following this, we met with three representatives of the group to discuss our proposed edits and additional items. Lastly, these edits were considered by the group before a final list of items was shared with us. A five-point Likert-type scale (1 = *strongly disagree* to 5 = *strongly agree*) with an additional *not applicable* option was selected for the response scale. Some items were reverse keyed (see Appendix A). Total scores were averaged, so that participants' final monotropism score was on a scale ranging from one to five, with higher scores indicating higher levels of monotropism.

Short Ritvo Autism and Asperger Diagnostic Scale (RAADS-14)

The shortened version of the Ritvo Autism and Asperger Diagnostic Scale (RAADS-14; Eriksson et al., 2013) is a screening instrument used to identify people with undiagnosed autism. Participants read 14 statements and select the answer which most closely reflects their own experience. Response options are: *True now and when I was young*, *True only now*, *True when I was young*, or *Never true*. Here, *When I was young* referred to the age of 17 years or younger. The measure was scored according to the published guide, with higher scores indicating a higher likelihood of autism (minimum score: 0, maximum score: 42).

Short Autism Spectrum Quotient (AQ-10)

The shortened version of the Autism Spectrum Quotient (AQ-10; Allison et al., 2013) measures the expression of traits related to autism within the general population. Participants read 10 statements and responded on a four-item Likert scale (1 = *Definitely agree* to 4 = *Definitely disagree*). The measure was scored according to the published guide, with higher scores indicating a higher expression of autism-related traits (minimum score: 0, maximum score: 10).

Procedure

Participants completed the survey online via a Qualtrics webpage after receiving information about the study and providing informed consent. Within each questionnaire, all items were presented in a randomised order. At the end of the questionnaire, participants were invited to provide feedback on the monotropism questionnaire and monotropism as a theory of autism. These qualitative data were collected to inform future research plans and were not analysed in this study. After our analyses were completed, the results were discussed with the aforementioned autistic-led community group.

Analysis

All analyses were conducted using R 4.2.0 (R Core Team, 2022; see Appendix D for packages) and were pre-registered at <https://osf.io/g2nm7>. Participants who are autistic (diagnosed by a clinician or self-identified) or who reported that they “think they might be autistic” were classified as being part of the autism group for this study, reflecting that not all individuals who may be autistic will access clinical diagnosis. Participants who were not autistic or described themselves as “not sure” about their autism status were categorised as non-autistic. The same categorisation pattern was applied to ADHD status (see Table 1 for details).

Following basic descriptive characterisation of the data, linear regressions were used to compare monotropism scores across groups and test the interaction between ADHD and autism status. Pearson correlations were used to assess patterns of correlation between the monotropism scores and the scales (and sub-scales) of the pre-existing autism measures. For the autistic sample, we used regression to conduct a sensitivity analysis of how monotropism scores varied with specific diagnostic status (diagnosed, self-identified, “might be autistic”).

An exploratory factor analysis (EFA) was conducted to explore the dimensionality of the monotropism measure and inform whether the items should be organised into sub-scales, as well as to inform future scale refinements. Factor retention was guided by a parallel analysis of principal components, a minimum average partial (MAP) test, visual inspection of a scree plot and conceptual and pragmatic considerations such as factor interpretability and determination (e.g., where minor factors that were unlikely to measure a sub-dimension reliably arose, we considered whether a factor solution with a smaller number of dimensions was optimal). Once the optimal number of factors was selected, a minimum residuals (minres) approach was used for factor extraction and oblique (oblimin) factor rotation applied for multi-dimensional solutions. Internal consistency reliability was assessed by McDonald’s (1999) omega, with $>.70$ indicating acceptable reliability.

Results

Group comparisons

A Mann-Whitney U test was used to compare autistic and non-autistic participants as parametric assumptions were not met for these data. The mean monotropism score of autistic

participants ($M = 4.15$, $SD = .347$) was significantly higher than for non-autistic participants ($M = 3.19$, $SD = .578$), $W = 21220$, $p < .001$ (see Figure 1).

Mean monotropism scores by autism and ADHD status together are shown in Figure 2. A multiple linear regression (which met model assumptions) with monotropism scores regressed on ADHD and autism status and their interaction explained a significant proportion of the variance in monotropism scores [$F(3, 1105) = 497$, $p < .001$, $R^2 = .574$, $R^2_{adjusted} = .573$]. Parameter estimates are provided in Table 2. These show that compared to having neither ADHD nor being autistic, having ADHD but not being autistic [$\beta = 0.55$, $t(1105) = 11.54$, $p < .001$], being autistic but not having ADHD [$\beta = 1.04$, $t(1105) = 30.52$, $p < .001$] and the combination of ADHD and autism [$\beta = -0.43$, $t(1105) = -7.51$, $p < .001$] significantly predicted monotropism scores. These findings suggest that both ADHD and autism are associated with higher monotropism; however, their effects are interactive such that monotropism scores are not as high for autistic individuals with ADHD as would be expected based on purely additive effects.

In a sensitivity analysis, having a clinical autism diagnosis (intercept) significantly predicted monotropism scores, $\beta = 4.18504$, $t(753) = 245.06$, $p < .001$, with a mean score of 4.19 ($SD = .352$) being observed in this participant group. Participants who self-identified as autistic had the same mean score of 4.19 ($SD = .319$), so this diagnostic status did not significantly predict a change in monotropism scores, $\beta = 0.00846$, $t(753) = .29$, $p = .772$. Finally, participants who thought they might be autistic had lower mean monotropism scores than the other autism groups ($M = 4.01$, $SD = .329$) and their diagnostic status significantly predicted a decrease in monotropism scores, $\beta = -0.18003$, $t(753) = -5.64$, $p < .001$.

Convergent Validity

Spearman's correlation coefficients suggested a strong positive correlation between participants' total monotropism scores and total RAADS-14 scores, $r_s = .78$, $p < .00$ and subscale scores for mentalising deficits ($r_s = .742$, $p < .001$), sensory reactivity ($r_s = .686$, $p < 0.001$), and social anxiety ($r_s = .616$, $p < 0.001$). Total monotropism scores were also strongly correlated with total AQ-10 scores, $r_s = .71$, $p < .001$. These are illustrated in Supplementary Figures 1 and 2.

Exploratory Factor Analysis

A parallel analysis of the monotropism item scores suggested retaining eight factors, while a MAP test suggested retaining four factors, and the scree plot suggested between four and eight factors. Therefore, five separate EFAs solutions were fitted and inspected. Apart from the Bayesian information criterion, which had the closest fit for the seven-factor structure, model fit indices supported the 8-factor model, as shown in Table S2 of Supplementary Materials. A high degree of conceptual clarity was observed for the eight-factor structure and it made conceptually important distinctions that were obscured in other solutions (see Supplementary Materials for details). Considering this together with the evidence from the parallel analysis, scree plot, and fit statistics, we considered an eight-factor structure to provide the optimal representation of the structure of the monotropism questionnaire. In interpreting the chosen factor solution, factor loadings $> |.30|$ were considered salient (Hair et al., 1998). Item 26 loaded significantly ($> .30$) on factor four and eight, but was retained and assigned to the former factor, as the loading was slightly higher. Meanwhile, items 7, 11, 17, 24, 29, and 41 did not load strongly on any factor ($< .30$) and were excluded. Factor loadings and dimension themes for the eight-factor minres solution are provided in Table S3 and Figure 3.

Internal Consistency

McDonald's omega hierarchical and total (McDonald, 1999; Revelle & Zinbarg, 2009) were calculated for the eight-factor structure of the monotropism questionnaire, excluding the items which were excluded by the eight-factor EFA, to assess the measure's internal consistency. The monotropism questionnaire showed a high level of internal consistency, $\omega_t = .96$, $\omega_h = .81$ for the general factor.

Discussion

This study presents the first systematic attempt to develop and validate a novel self-report measure of monotropism in autistic and non-autistic people, also taking account of self-reported ADHD status. The results support our hypothesis that mean monotropism questionnaire scores are significantly higher for autistic compared to non-autistic participants. Participants' ADHD status, autism status, and their interaction significantly predict their mean monotropism questionnaire scores, meaning that autism and ADHD are independently and jointly associated with higher self-report monotropism. Furthermore, the monotropism measure displays good psychometric properties. Total monotropism scores are positively

correlated with RAADS-14 and AQ-10 scores, indicating that the measure has good convergent validity. An exploratory factor analysis indicates that a 41-item version of the monotropism questionnaire assesses eight latent dimensions and has high internal consistency.

When conceptualising Monotropism as a theory of autism, Murray and colleagues (2005) predicted that high levels of monotropism would “pick out” autistic people, now empirically demonstrated for the first time in our results. That said, the non-autistic group distribution is flatter and has a larger range, spread across the full continuum of possible monotropism scores. Meanwhile, the autistic group distribution is more tightly packed, with only a handful of autistic participants scoring lower than the mean score of the non-autistic group. This difference in distributions suggests that monotropism may be normally distributed in the non-autistic population while also being highly concentrated amongst autistic people. It is possible that some of the non-autistic group who scored highly on the monotropism questionnaire might have undiagnosed autism, but this is unlikely given our inclusive approach to classifying our autistic sample in this study.

Additionally, our results indicate that ADHD status has an effect on participants’ monotropism scores. As mentioned previously, there is no research that specifically links Monotropism theory to ADHD. However, there are indications that ADHD and autism share some attentional features such as hyperfocus (e.g., Grotewiel et al., 2022), which is conceptually very similar to the monotropic “attention tunnel” – namely the absorbing experience of focused attention, often to the exclusion of all else. It may be that monotropic attention is not only captured endogenously by individuals’ special interests but also exogenously by highly salient environmental stimuli - as has been observed in autistic children (Amso et al., 2014). This may provide the basis for an elaboration of Monotropism that could allow the theory to account for the co-occurrence of autism and ADHD.

The subscales identified in the monotropism measure align with many established features of autism. Factors one and seven capture the phenomenon of deep interests in specific topics that can persist throughout life and which can generate strong feelings of joy and relaxation (Spiker et al., 2012). Factors four and five relate to the monotropic “attention tunnel”. The former concerns the disruptive effects of unexpected or sensorily demanding stimuli when focused (Trembath et al., 2012) and, conversely, the positive effects of quiet and predictable environments (Gomot & Wicker, 2012). The latter describes external events, time, and interoception going unnoticed during periods of focus (Ashinoff & Abu-Akel, 2021).

Other factors are more focused on anxiety-related aspects of autism. Factor two captures autistic people's propensity for rumination and anxiety (Jenkinson et al., 2020). Another well-documented aspect of autism linked to anxiety is adherence to routines and rituals (Ozsivadijan et al., 2021), as captured by factor three. Factor six corresponds to accounts of autistic people finding decision-making difficult (Luke et al., 2011) and to the concept of autistic inertia (Buckle et al., 2021). Finally, factor eight describes differences in social communication and interaction, as captured in the diagnostic criteria (APA, 2013).

Although these results support the validity of the monotropism questionnaire, there are limitations to be addressed by future research. The participants in the present study were predominantly White and highly educated, as were the autistic contributors to item-generation. Additionally, although this study used relatively accessible self-report measures, participation may have been inaccessible to autistic people with co-occurring intellectual disabilities. Therefore, the generalisability of our findings cannot be established. Future work should also examine convergent validity with measures of other theoretically related constructs, such as hyperfocus (Grotewiel et al., 2022) and how interest impacts on learning. The test-retest reliability of the monotropism questionnaire was not assessed in this study but is an important area for future research. Finally, we hope in due course to present an analysis of qualitative views on the utility of the monotropism model for autistic people, and integrate this with the burgeoning qualitative evidence base for monotropism.

In conclusion, this research provides a promising and important step in the investigation of the predictions of Monotropism theory. A combination of participatory methods and robust psychometric evaluation against established instruments gives confidence in these data. Further empirical investigation of Monotropism could reframe our understanding of autism and inform how we support autistic people.

Acknowledgements

We thank all the members of the Interest Systems Group for their invaluable contribution to measure development and recruitment.

Research in context

Evidence before this study The authors drew on all published papers explicitly articulating the Monotropism theory of autism, and triangulated them with a targeted account of relevant literature on autistic cognition that aligns with the predictions of this theory. Since this paper

attempts the first empirical measure of monotropism there is no quantitative literature directly examining the theory.

Added value of this study This study provides the first quantitative examination of monotropism in a broadly-defined autistic sample, encompassing clinically diagnosed and self-identified people. We also capture self-reported ADHD status to give clues to the future application of Monotropism theory in research and practice.

Implications of all the available evidence The research indicates that monotropism is a shared characteristic amongst autistic people, and relatively rare in the non-autistic population. We also share preliminary evidence that monotropism scores are higher in people with ADHD. This initial evidence on an as-yet-untested theory can motivate further enquiry and future implementation.

References

- Allison, C., Auyeung, B., & Baron-Cohen, S. (2012). Toward brief “red flags” for autism screening: the short autism spectrum quotient and the short quantitative checklist in 1,000 cases and 3,000 controls. *Journal of the American Academy of Child & Adolescent Psychiatry*, 51(2), 202-212. <https://doi.org/10.1016/j.jaac.2011.11.003>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Association. <https://doi.org/10.1176/appi.books.9780890425596>
- Amso, D., Haas, S., Tenenbaum, E., Markant, J., & Sheinkopf, S. J. (2014). Bottom-up attention orienting in young children with autism. *Journal of autism and developmental disorders*, 44(3), 664-673. <https://doi.org/10.1007/s10803-013-1925-5>
- Ashinoff, B. K., & Abu-Akel, A. (2021). Hyperfocus: The forgotten frontier of attention. *Psychological Research*, 85(1), 1-19. <https://doi.org/10.1007/s00426-019-01245-8>
- Bertilsdottir Rosqvist, H. (2019). Doing things together: Exploring meanings of different forms of sociality among autistic people in an autistic work space. *Alter*, 13(3), 168-178.
- Bertilsdotter Rosqvist, H., Hultmann, L., Wiklund, S.Ö., Nygren, A., Storm, P. & Sandberg, G. (2023). Intensity and Variable Attention: Counter Narrating ADHD, from ADHD Deficits to ADHD Difference, *The British Journal of Social Work*, bcad138, <https://doi.org/10.1093/bjsw/bcad138>

- Bottema-Beutel, K., Kapp, S. K., Lester, J. N., Sasson, N. J., & Hand, B. N. (2021). Avoiding Ableist Language: Suggestions for Autism Researchers. *Autism in Adulthood*, 3(1), 18–29. <https://doi.org/10.1089/aut.2020.0014>
- Buckle, K. L., Leadbitter, K., Poliakoff, E., & Gowen, E. (2021). “No way out except from external intervention”: First-hand accounts of autistic inertia. *Frontiers in Psychology*, 1592. <https://doi.org/10.3389/fpsyg.2021.631596>
- Collignon, O., Charbonneau, G., Peters, F., Nassim, M., Lassonde, M., Lepore, F., Mottron, L., & Bertone, A. (2013). Reduced multisensory facilitation in persons with autism. *Cortex*, 49(6), 1704–1710. <https://doi.org/10.1016/j.cortex.2012.06.001>
- Craig, F., Margari, F., Legrottaglie, A. R., Palumbi, R., de Giambattista, C., & Margari, L. (2016). A review of executive function deficits in autism spectrum disorder and attention-deficit/hyperactivity disorder. *Neuropsychiatric disease and treatment*, 12, 1191–1202. <https://doi.org/10.2147/NDT.S104620>
- Eriksson, J. M., Andersen, L. M., & Bejerot, S. (2013). RAADS-14 Screen: validity of a screening tool for autism spectrum disorder in an adult psychiatric population. *Molecular Autism*, 4(1), 1–11. <https://doi.org/10.1186/2040-2392-4-49>
- Fletcher-Watson, S., Adams, J., Brook, K., Charman, T., Crane, L., Cusack, J., Leekam, S., Milton, D., Parr, J. R., & Pellicano, E. (2019). Making the future together: Shaping autism research through meaningful participation. *Autism*, 23(4), 943–953. <https://doi.org/10.1177/1362361318786721>
- Goldstein, G., Johnson, C. R., & Minshew, N. J. (2001). Attentional processes in autism. *Journal of autism and developmental disorders*, 31(4), 433–440. <https://doi.org/10.1023/A:1010620820786>
- Gomot, M., & Wicker, B. (2012). A challenging, unpredictable world for people with autism spectrum disorder. *International Journal of Psychophysiology*, 83(2), 240–247. <https://doi.org/10.1016/j.ijpsycho.2011.09.017>
- Grotewiel, M. M., Crenshaw, M. E., Dorsey, A., & Street, E. (2022). Experiences of hyperfocus and flow in college students with and without Attention Deficit Hyperactivity Disorder (ADHD). *Current Psychology*, 1, 1–11. <https://doi.org/10.1007/s12144-021-02539-0>
- Hair, J. F., Anderson, R. E., Tatham, R. L., & Black, W. C. (1998), *Multivariate data analysis* (5th ed.). Prentice Hall.
- Happé, F., & Frith, U. (2020). Annual Research Review: Looking back to look forward – changes in the concept of autism and implications for future research. *Journal of Child*

- Psychology and Psychiatry and Allied Disciplines*, 61(3), 218-232.
<https://doi.org/10.1111/jcpp.13176>
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural equation modeling: a multidisciplinary journal*, 6(1), 1-55. <https://doi.org/10.1080/10705519909540118>
- Jenkinson, R., Milne, E., & Thompson, A. (2020). The relationship between intolerance of uncertainty and anxiety in autism: A systematic literature review and meta-analysis. *Autism*, 24(8), 1933-1944. <https://doi.org/10.1177/1362361320932437>
- Lawson, W. (2010) *The Passionate Mind: How People with Autism Learn*. Jessica Kingsley Publishers, UK.
- Leatherland, J. (2018). *Understanding how autistic pupils experience secondary school: Autism criteria, theory and FAME*. PhD Thesis, Sheffield Hallam University, (United Kingdom) ProQuest Dissertations Publishing
- Luke, L., Clare, I. C. H., Ring, H., Redley, M., & Watson, P. (2012). Decision-making difficulties experienced by adults with autism spectrum conditions. *Autism*, 16(6), 612-621. <https://doi.org/10.1177/1362361311415876>
- McDonald, R.P. (1999). *Test theory: A unified treatment*. Erlbaum.
- McDonnell, A., & Milton, D. (2014). Going with the flow: reconsidering ‘repetitive behaviour’ through the concept of ‘flow states’. In: Jones, G. & Hurley, E. Eds. *Good Autism Practice: autism, happiness and wellbeing*. BILD, Birmingham, UK, pp. 38-47.
- Milton, D. E. (2014). Autistic expertise: A critical reflection on the production of knowledge in autism studies. *Autism*, 18(7), 794-802. <https://doi.org/10.1177/1362361314525281>
- Murray, D., Lesser, M., & Lawson, W. (2005). Attention, monotropism and the diagnostic criteria for autism. *Autism*, 9(2), 139-156. <https://doi.org/10.1177/1362361305051398>
- Murray, M. J. (2010). Attention-deficit/hyperactivity disorder in the context of autism spectrum disorders. *Current psychiatry reports*, 12, 382-388.
- Murray, F. (2018) Me and monotropism: a unified theory of autism. *The Psychologist*, downloaded from <https://www.bps.org.uk/psychologist/me-and-monotropism-unified-theory-autism>
- Murray, D. (2020). Monotropism—an interest based account of autism. *Encyclopedia of autism spectrum disorders*, 10, 978-981.
- Ozsivadjian, A., Hollocks, M. J., Magiati, I., Happé, F., Baird, G., & Absoud, M. (2021). Is cognitive inflexibility a missing link? The role of cognitive inflexibility, alexithymia and intolerance of uncertainty in externalising and internalising behaviours in young people

- with autism spectrum disorder. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 62(6), 715-724. <https://doi.org/10.1111/jcpp.13295>
- Revelle, W., & Zinbarg, R. E. (2009). Coefficients alpha, beta, omega, and the glb: Comments on Sijtsma. *Psychometrika*, 74(1), 145-154. <https://doi.org/10.1007/s11336-008-9102-z>
- Rong, Y., Yang, C. J., Jin, Y., & Wang, Y. (2021). Prevalence of attention-deficit/hyperactivity disorder in individuals with autism spectrum disorder: a meta-analysis. *Research in Autism Spectrum Disorders*, 83, 101759. <https://doi.org/10.1016/j.rasd.2021.101759>
- Rumball, F., Brook, L., Happé, F., & Karl, A. (2021). Heightened risk of posttraumatic stress disorder in adults with autism spectrum disorder: The role of cumulative trauma and memory deficits. *Research in Developmental Disabilities*, 110, 103848.
- Schwarz, G. (1978). Estimating the dimension of a model. *The annals of statistics*, 461-464. <http://www.jstor.org/stable/2958889>
- Spiker, M. A., Lin, C. E., Van Dyke, M., & Wood, J. J. (2012). Restricted interests and anxiety in children with autism. *Autism*, 16(3), 306-320. <https://doi.org/10.1177/1362361311401763>
- Steiger, J. H., & Lind, J. C. (1980, May). *Statistically based tests for the number of factors* [Conference presentation]. Annual spring meeting of the Psychometric Society, Iowa City, IA.
- Tucker, L. R., & Lewis, C. (1973). A reliability coefficient for maximum likelihood factor analysis. *Psychometrika*, 38(1), 1-10. <https://doi.org/10.1007/BF02291170>
- Wood, R. (2021). Autism, intense interests and support in school: from wasted efforts to shared understandings, *Educational Review*, 73(1), 34-54.
- Woods, R. (2019). An updated interest-based account (monotropism theory) and a demand avoidance phenomenon discussion. *Disability Research Forum, Sheffield Hallam University*, doi: [10.13140/RG.2.2.32443.46884/1](https://doi.org/10.13140/RG.2.2.32443.46884/1)

Tables

Table 1: *Diagnostic status of the sample and groupings for analysis*

Demographic Characteristic		Count (%)		
		Autistic	Non-Autistic	Total
ADHD status				
ADHD (clinical diagnosis)		174 (23.02%)	43 (12.15%)	217 (19.55%)
ADHD (self-identified)	ADHD group	73 (9.66%)	6 (1.69%)	79 (7.12%)
I think I may have ADHD		186 (24.60%)	51 (14.41%)	237 (21.35%)
Unsure		124 (16.40%)	54 (15.25%)	178 (16.04%)
No ADHD	No ADHD group	199 (26.32%)	199 (56.21%)	398 (35.86%)
No information		0 (0%)	1 (0.28%)	1 (0.09%)
Autism status				
Autistic (clinical diagnosis)		394 (52.12%)	-	394 (35.50%)
Autistic (self-identified)	Autistic group	204 (26.98%)	-	204 (18.38%)
I think I may be autistic		158 (20.90%)	-	158 (14.23%)
Unsure		-	87 (24.58%)	87 (7.84%)
Not autistic	Non-autistic group	-	267 (75.42%)	267 (24.05%)

Table 2: Multiple linear regression table of the model used to predict the effect of ADHD status and autism status on mean monotropism questionnaire scores

Predictor	Mean Monotropism Questionnaire Score		
	Estimate	95% CI	P
Intercept	3.03	2.98 – 3.08	< .001
Autism	1.04	0.98 – 1.11	< .001
ADHD	0.56	0.46 – 0.65	< .001
Autism*ADHD	-0.43	-0.54 – -0.31	< .001
Observations	1109		
R ² / R ² _{adjusted}	0.574 / 0.573		

Note. CI = confidence interval.

Figures

Figure 1: Mean monotropism questionnaire scores by autism status

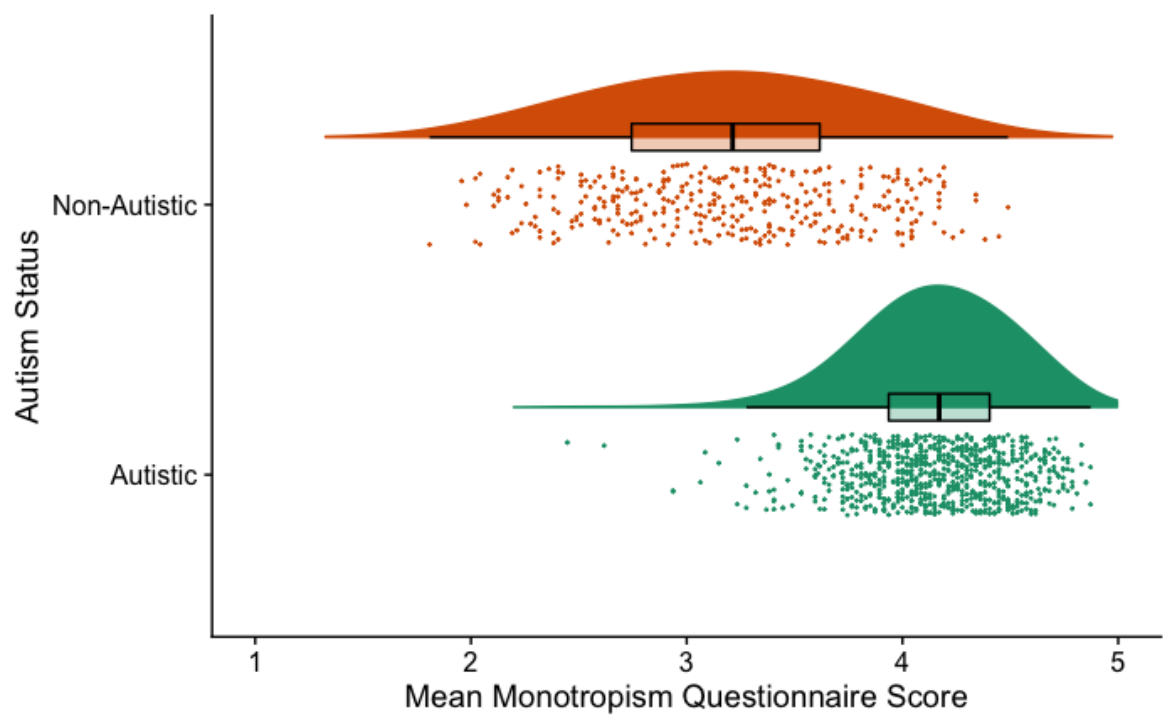
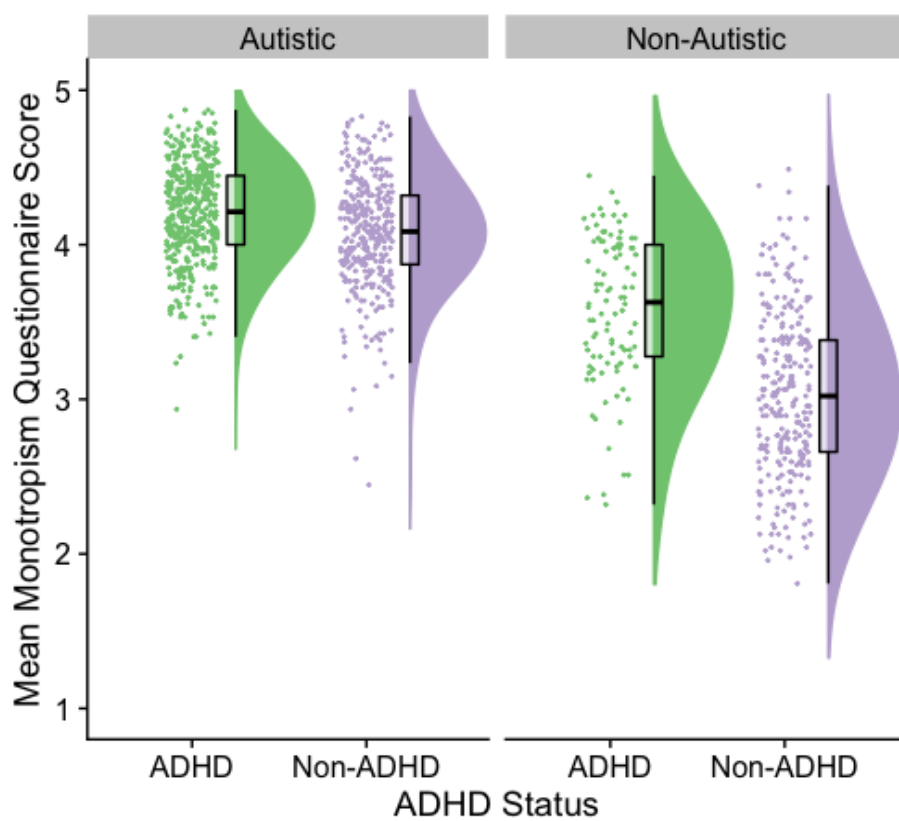


Figure 2: Mean monotropism questionnaire scores by autism status and ADHD status



Supplementary Table S1: *Demographic characteristics of the sample*

Demographic Characteristic Count (%)	Autistic	Non-Autistic	Whole Sample
<i>Gender</i>			
Male	128 (16.93%)	74 (20.90%)	202 (18.20%)
Female	439 (58.07%)	258 (72.88%)	697 (62.79%)
Non-binary / third gender	177 (23.41%)	19 (5.37%)	196 (17.66%)
Prefer not to say / no information	12 (1.59%)	3 (0.85%)	15 (1.35%)
<i>Ethnicity</i>			
White	668 (88.36%)	313 (88.42%)	981 (88.38%)
Black / African / Caribbean / Black British	8 (1.06%)	9 (2.54%)	17 (1.53%)
Asian / Asian British	21 (2.78%)	12 (3.39%)	33 (2.97%)
Mixed / multiple ethnic groups	41 (5.42%)	13 (3.67%)	54 (4.87%)
Other ethnicity / no information	18 (2.38%)	7 (1.98%)	25 (2.25%)
<i>Highest level of education</i>			
A-level or equivalent and below	104 (13.76%)	28 (7.91%)	132 (11.89%)
Vocational or college qualification	75 (9.92%)	25 (7.06%)	100 (9.01%)
Undergraduate degree	224 (29.63%)	91 (25.71%)	315 (28.38%)
PG degree / professional qualification	342 (45.24%)	210 (59.32%)	552 (49.73%)
No information	11 (1.46%)	0 (0%)	11 (0.99%)
<i>ADHD status</i>			
ADHD (clinical diagnosis)	174 (23.02%)	43 (12.15%)	217 (19.55%)
ADHD (self-identified)	73 (9.66%)	6 (1.69%)	79 (7.12%)
I think I may have ADHD	186 (24.60%)	51 (14.41%)	237 (21.35%)
Unsure	124 (16.40%)	54 (15.25%)	178 (16.04%)
No ADHD	199 (26.32%)	199 (56.21%)	398 (35.86%)
No information	0 (0%)	1 (0.28%)	1 (0.09%)
<i>Autism status</i>			
Autistic (clinical diagnosis)	394 (52.12%)	-	394 (35.50%)
Autistic (self-identified)	204 (26.98%)	-	204 (18.38%)
I think I may be autistic	158 (20.90%)	-	158 (14.23%)
Unsure	-	87 (24.58%)	87 (7.84%)
Not autistic	-	267 (75.42%)	267 (24.05%)

Supplementary Table S2: *Comparison of exploratory factor analyses models' fit indices*

Model	Fit Indices			
	RMSR	RMSEA	TLI	BIC
4 factors	.03	0.049	0.874	-3040
5 factors	.03	0.044	.897	-3293
6 factors	.02	.04	.914	-3428
7 factors	.02	.037	.927	-3464
8 factors	.02	.035	.935	-3404

Note. Best fit values in line with Hu and Bentler's (1999) guidelines are in bold. Root mean square of residuals (RMSR) values closest to 0 indicate better model fit. Root Mean Square Error of Approximation (RMSEA; Steiger & Lind, 1980) index values $\leq .05$ indicate very good model fit. Tucker-Lewis Index (TLI; Tucker & Lewis, 1973) values $\geq .90$ indicate good model fit. Smaller Bayesian information criterion (BIC; Schwarz, 1978) values indicate better model fit.

Supplementary Table S3: Main and cross-loadings of the items of the monotropism questionnaire, dimension themes, and percentage of variance explained by each factor for the eight factor model

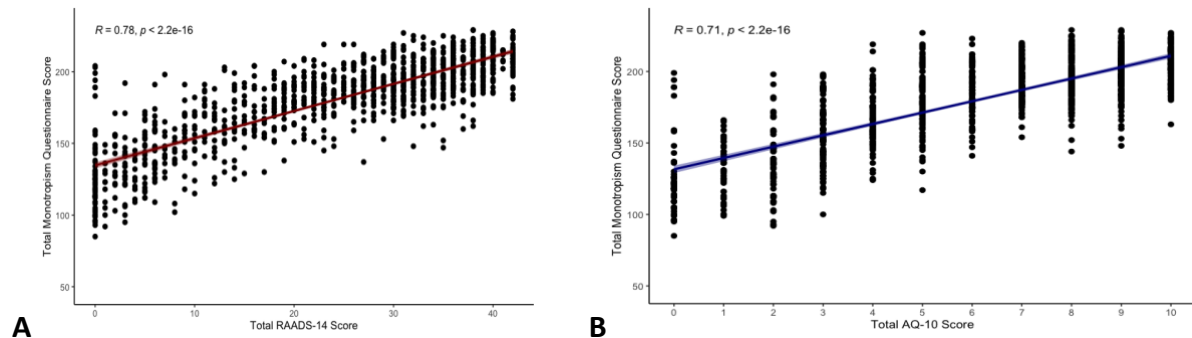
Item	Factor								Dimension (% of variance explained)
	1	2	3	4	5	6	7	8	
44	.565	.112		.110		-.127			Special interests (15%)
43	.551						.168		
25	.458						.133		
45	.396							.164	
28	.391			-.114		.130	.133	.197	
39	.378	.123				.170		.123	Rumination and anxiety (12%)
40	.318	.243		.161	.115				
14	.314	-.146		.133			.206	.104	
37		.635				.116	.160		
42	.127	.523				.295			
22	-.175	.516		.204			.153		Need for routines (10%)
36		.458					.153		
47	.264	.435		.122					
46		.358		.157	.108				
20			.884						
21			.838						Environmental impact on the attention tunnel (17%)
4	.102	.149		.610					
2			.134	.605					
3				.595		.101			
16	.108			.487	.130			.166	

cont.

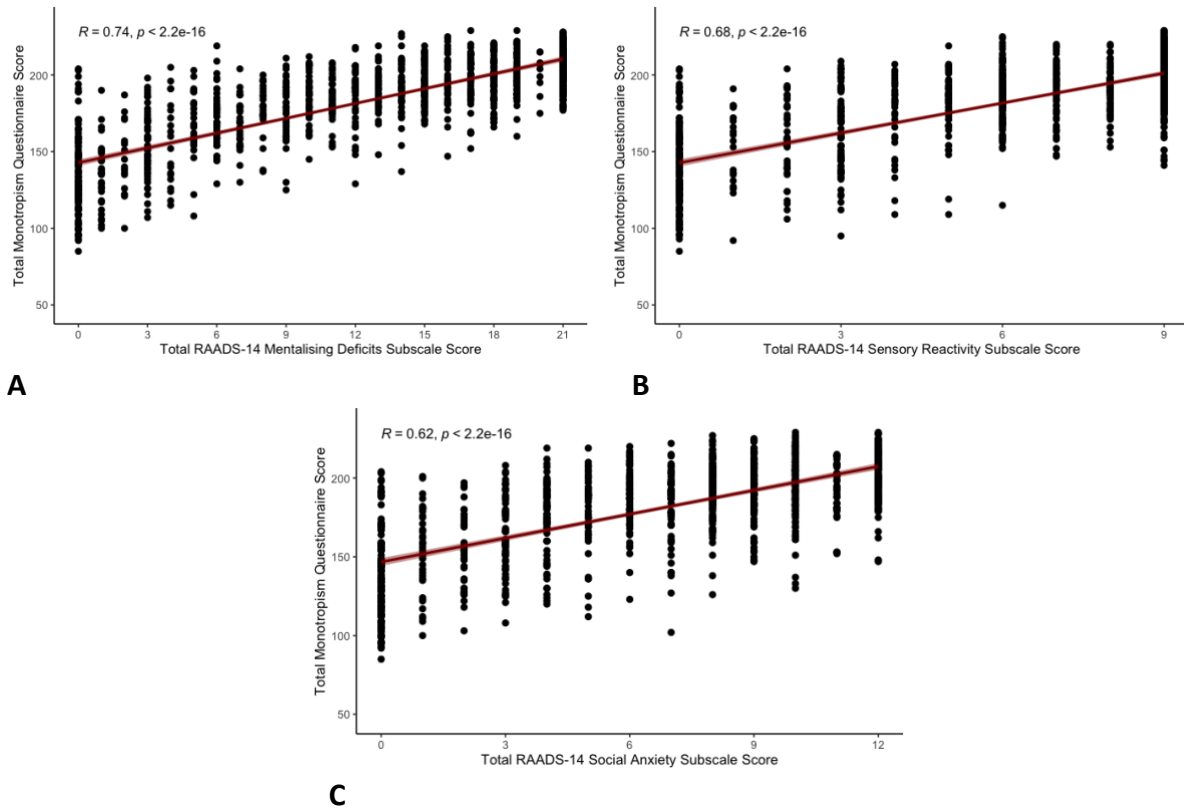
Item	Factor								Dimension (% of variance explained)
	1	2	3	4	5	6	7	8	
5	.138			.459			.108		
26				.368	.103			.317	
1			.239	.365					
18	.120			.326	.173	.190			
32					.932				Losing track of other factors when focusing on special interests (13%)
33					.773				
12	.140			.111	.422	.160	.189	-.113	
10	.283	-.115			.311	.181	.211		
34		.150				.636			Struggle with decision- making (11%)
35		.116			.106	.576		.130	
38	.101					.447			
23	.119	-.124	-.142	.274		.322	.169		
8						-.100	.678		Anxiety-reducing effect of special interests (9%)
19		.182	.101				.484		
9	.117		.113			.143	.339		
13						.171	.156	.478	Managing social interactions (13%)
15				.149	.119			.475	
31		-.104		.245				.461	
27	.211				.107			.382	
6		.101					.114	.329	
30	.248	.231				.109		.304	

Note. Factor loadings > .30 are in bold.

Supplementary Figure 1: (A) Scatterplot with a linear regression line and confidence interval band showing the correlation between total monotropism scores and total RAADS-14 scores. (B) Scatterplot with a linear regression line and confidence interval band showing the correlation between total monotropism scores and total AQ-10 scores.



Supplementary Figure 2: Scatterplots with a linear regression line and confidence interval band showing the correlation between total monotropism scores and sub-scales for the RAADS-14 pertaining to (A) Mentalizing Deficits, (B) Sensory Reactivity and (C) Social Anxiety.



Supplementary Materials: the 8 Factor Structure of the Monotropism Questionnaire**Factor 1: Special interests**

- 44) When I am interested in a topic, I like to learn everything I can about that topic.
- 43) When I am interested in something, I tend to be passionate about it.
- 25) I am usually passionate about a few topics at any one time in my life.
- 45) I am still fascinated by many of the things I was interested in when I was much younger.
- 28) I often engage in lengthy discussions on topics I find interesting even though my conversational partner(s) do not.
- 39) People tell me I get fixated on things.
- 40) I find a problem I can't solve distressing and/or hard to put down.
- 14) I tend to do activities because I find them interesting, instead of due to societal expectations.

Factor 2: Rumination and anxiety

- 37) I sometimes become highly anxious by focusing on the many possible situations that might occur at a future event.
- 42) I often get stuck thinking about all the possibilities that might come out of a decision.
- 22) I often experience anxiety over matters I have little certainty over.
- 36) I sometimes focus on an incident for a substantial time (days) after the event.
- 47) I often loop back to previous thoughts.
- 46) I rarely find myself getting stuck in loops of thought. *

Factor 3: Need for routines

- 20) Routines provide an important source of stability and safety.
- 21) I manage uncertainty by creating routines.

Factor 4: Environmental impact on the attention tunnel

- 4) I find sudden unexpected disruptions to my attention startling.
- 2) I need a quiet and predictable environment for me to switch from one task to another easily.
- 3) I often struggle to concentrate in busy and/or unpredictable environments.
- 16) I don't mind if someone interrupts me when I'm in the middle of an activity. *
- 5) It's distressing to be unexpectedly pulled away from something I'm engaged in.
- 26) I have trouble filtering out sounds when I am not doing something I'm focused on.
- 1) After a period of instability, I need a quiet and predictable environment.
- 18) I often find it difficult to switch topics after engaging in an activity for a long time.

Factor 5: Losing track of other factors when focusing on special interests

- 32) Often when I am focused on activities, I do not notice I am thirsty or hungry.
- 33) Often when I am focused on activities, I do not notice I need the bathroom.
- 12) I often lose sense of time when engaging in activities I am passionate about.
- 10) I am often totally focused on activities I am passionate about, to the point I am unaware of other events.

Factor 6: Struggle with decision-making

- 34) When there is a lot of information to consider, I often struggle to make a decision.
- 35) Sometimes making a decision is so hard I get physically stuck.
- 38) Sometimes when I am focused on an activity, I do not recall all the information I might need to make good decisions.
- 23) I find it difficult to engage in a task of no interest to me even if it is important.

Factor 7: Anxiety-reducing effect of special interests

- 8) Involvement in an activity of interest often reduces my anxiety level.
- 19) I often engage in activities I am passionate about to escape from anxiety.
- 9) I find social interactions more comfortable if communicating about a topic of interest to me.

Factor 8: Managing social interactions

- 30) I can sometimes be very distressed by a topic that others think of as trivial.
- 6) I rarely find simultaneously holding eye contact and making a verbal conversation with another person uncomfortable. *
- 27) I usually mean what I say and no more than that.
- 31) I find it easy to keep up with group discussions where everyone is speaking. *
- 15) I rarely find social situations chaotic. *
- 13) I sometimes avoid talking because I cannot reliably predict how others will react, especially strangers.

No factor

- 11) I can get quite good at something even if I'm not especially interested in it. *
- 17) When I'm working on something, I'm open to helpful suggestions.*
- 7) I often notice details that others do not.
- 41) I tend to feel quite self-conscious unless I'm deeply absorbed in a task.
- 29) I sometimes accidentally say something others find offensive/ rude when I am focused on a task.
- 24) I often find engaging in stimming (e.g., fidgeting, rocking) to be relaxing.

Note. Reverse scored items are marked with *.